
Where Does Latent Planning Fail?

Controlled Bottleneck Attribution for World-Model Planning

Anonymous Author(s)

Affiliation

Address

email

Abstract

When a latent world-model planner selects a poor action, inaccurate prediction is a natural suspect, but decisions also depend on how predicted futures are scored and which candidates search exposes. We introduce a controlled attribution protocol that changes one interface at a time and executes every selected action in the simulator. Candidate-matched oracle substitutions estimate recoverable prediction, decision-metric, and search headroom using initial states as the independent units. A controlled local Push-T protocol exhibits negligible prediction headroom but a substantial metric gap; the diagnosis-selected scorer reduces normalized regret by 0.0815 (95% CI [0.0563, 0.1079]). An independent Wall replication reverses the diagnosis: prediction headroom is 0.5636 [0.4871, 0.6373], and a pre-specified predictor repair improves regret by 0.3891 [0.2930, 0.4804], while a wrong-layer metric repair does not. Under the official high-dimensional Push-T H6 CEM planner, prediction headroom exceeds metric headroom, and changing the scorer also changes adaptive search coverage; its end-to-end improvement is directionally positive but statistically inconclusive. Thus, the dominant recoverable bottleneck can migrate across planning regimes. These results support a practical rule: diagnose where decision quality is lost before retraining the world model or changing the planner.

1 Introduction

Latent world models make planning computationally attractive by predicting future representations instead of reconstructing pixels [Hafner et al., 2019, Zhou et al., 2025, Assran et al., 2025, Terver et al., 2026]. A planner, however, contains more than a predictor. Candidate actions are proposed, future latents are predicted, a decision metric ranks them, and search selects an action for execution. A poor action can therefore arise even when predicted latents are close under the training loss. Conversely, visibly imperfect predictions can preserve enough information to select the best available action.

This distinction matters in practice. Predictor retraining may require new data and substantial compute. If recoverable decision quality is instead lost at the scoring or proposal interface, predictor improvements target the wrong component. The reverse error is equally possible: a task-aligned scorer cannot restore information that prediction has discarded. Generic prediction accuracy alone is therefore insufficient to localize a planning failure.

We ask: *where is the dominant recoverable bottleneck in a frozen world-model planning pipeline?* PhysPlan answers with controlled oracle substitutions. For the same initial state and candidate actions, it replaces predicted futures with encoded simulator futures, replaces latent distance with a frozen task readout, or compares queried actions with simulator oracles. Every resulting selection is evaluated

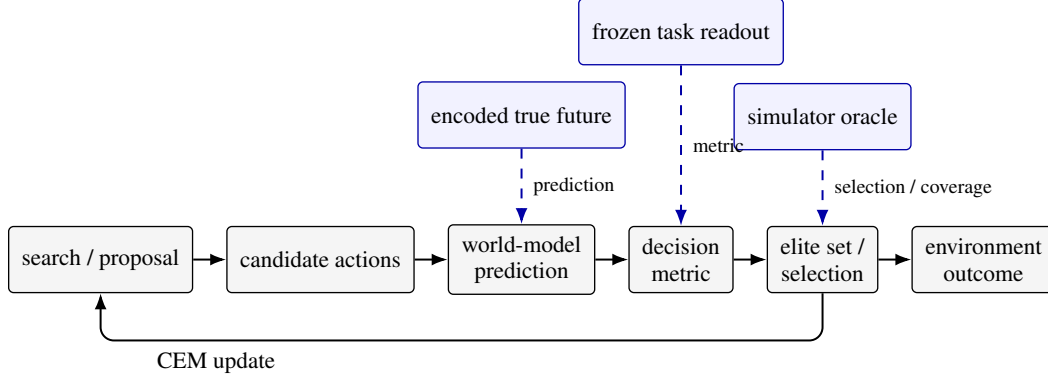


Figure 1: PhysPlan attributes recoverable decision headroom by controlled substitutions at planner interfaces. On a fixed candidate trace, encoded true futures isolate prediction headroom and a frozen task readout isolates metric headroom. Simulator oracles expose within-search selection and proposal coverage. Under adaptive CEM, elite selection updates the proposal, so a scorer intervention changes both ranking and future candidate coverage.

by simulator cost. These contrasts estimate the decision headroom recoverable at prediction, metric, proposal, and within-search selection interfaces (Figure 1); they are diagnostic contrasts, not an additive causal decomposition.

We test whether a diagnosis prospectively identifies a useful intervention. In a controlled local Push-T protocol, prediction headroom is nearly zero while decision-metric headroom is substantial; a pre-specified metric repair recovers almost exactly that gap, whereas increasing predictor inference precision does not recover planning performance. An independent Wall replication yields the opposite diagnosis, and its pre-specified predictor repair succeeds on unseen states while a wrong-layer metric repair does not. Finally, the official high-dimensional Push-T H6 planner reveals a regime shift: prediction headroom becomes larger than metric headroom, and the scorer changes the candidate distribution explored by adaptive CEM [Rubinstein, 1999]. The H6 scorer repair is directionally positive but statistically inconclusive.

Our contributions are threefold:

1. a candidate-matched, simulator-grounded protocol for estimating recoverable headroom at prediction, decision-metric, proposal, and selection interfaces of a frozen latent planner;
2. cross-regime evidence that controlled local Push-T is primarily metric limited, whereas an independent Wall replication is strongly prediction limited, together with diagnosis-selected interventions and task-appropriate wrong-layer controls; and
3. an official Push-T H6 study showing bottleneck migration and metric–search coupling: under adaptive CEM, changing the scorer changes both candidate ranking and future search coverage.

We do not propose a new world-model architecture, latent objective, or optimizer. The methodological claim is deliberately narrower: diagnose the dominant recoverable bottleneck before retraining or rescoreing.

2 Related Work

World models for control. Learned dynamics models support planning from pixels or compact state representations, from PlaNet and probabilistic ensemble control to recent scalable latent-control systems [Hafner et al., 2019, Chua et al., 2018, Hansen et al., 2024]. DINO-WM plans over frozen visual features without reconstructing observations [Zhou et al., 2025]; V-JEPA 2 and JEPA-WMs study action-conditioned joint-embedding prediction for physical planning [Assran et al., 2025, Terver et al., 2026]. These works establish strong model-and-planner pipelines. We instead hold a pipeline fixed and ask which interface offers recoverable decision headroom.

Prediction quality and decision utility. Value-aware loss functions, value-targeted regression, and value-equivalent models formalize that models should preserve information relevant to decisions rather than optimize generic one-step prediction alone [Farahmand et al., 2017, Ayoub et al., 2020, Grimm et al., 2020, Voelcker et al., 2025]. PhysPlan is compatible with this premise but addresses a different operational question: before changing a model or objective, can controlled substitutions diagnose whether prediction is presently the limiting interface? Our task readout is therefore an evaluation instrument and simple repair, not a claim that task-aligned scoring is new.

Sampling-based planning. CEM iteratively samples candidates, selects elites, and updates its proposal [Rubinstein, 1999]; this pattern is common in model-based control [Chua et al., 2018, Hafner et al., 2019]. A planner may fail because its queries never cover a good action or because it misranks actions it did query. Retaining every query and simulator-evaluating those candidates separates coverage from within-trace selection. It also exposes an important limit of fixed-trace attribution: once a scorer changes CEM elites, it changes the future proposal distribution as well as ranking.

3 Controlled Bottleneck Attribution

3.1 Planning pipeline and regret

For initial state s , a proposal mechanism supplies candidate action sequences $a \in A_s$. An encoder maps observations to latents; an action-conditioned predictor produces future latents $\hat{z}_s(a)$; and a decision metric $q(s, a)$ ranks candidates. The simulator cost $J_s(a)$, unavailable to the deployed planner, is used only for evaluation. Lower cost is better.

For method m , let $a_m(s) = \arg \min_{a \in A_s} q_m(s, a)$. On a finite candidate set, we report normalized regret

$$\mathcal{R}_m(s) = \frac{J_s(a_m) - \min_{a \in A_s} J_s(a)}{\max_{a \in A_s} J_s(a) - \min_{a \in A_s} J_s(a) + \epsilon}. \quad (1)$$

All headline uncertainty treats the cloned initial state (or H6 episode) as the independent unit. Multiple nearby candidates from one state are never treated as independent observations.

3.2 Candidate-matched oracle substitutions

The diagnostic compares four conditions while preserving the initial state, candidate actions, and simulator evaluation:

GT readout. A frozen task readout scores encoded simulator futures.

Predicted readout. The same readout scores world-model predictions.

Predicted L2. The baseline latent distance scores the same predictions.

Simulator oracle. Simulator cost selects the best candidate, either from the reference set or from actions queried by search.

The readout never observes the action; it maps latent transition information to simulator task cost and is fit only on development states. Thus it cannot exploit a direct action-to-cost shortcut.

Prediction and metric headroom are the paired state-level contrasts

$$G_{\text{pred}} = \mathbb{E}_s[\mathcal{R}_{\text{pred-readout}}(s) - \mathcal{R}_{\text{GT-readout}}(s)], \quad (2)$$

$$G_{\text{metric}} = \mathbb{E}_s[\mathcal{R}_{\text{pred-L2}}(s) - \mathcal{R}_{\text{pred-readout}}(s)]. \quad (3)$$

Proposal headroom compares the simulator-best queried candidate with the best reference candidate. Within-search selection headroom compares the planner’s selected queried candidate with the simulator-best queried candidate. These are recoverable contrasts under specific substitutions. They overlap, depend on the candidate set and readout capacity, and must not be summed or interpreted as a unique causal partition.

107 3.3 Prospective diagnosis and intervention

108 Before repair evaluation, each controlled protocol fixes the substitution definitions, dominance rule,
109 bootstrap, selected intervention, wrong-layer control, and success criterion. The diagnosed component
110 is the eligible gap whose lower confidence bound clears the frozen materiality margin and dominates
111 the alternatives; otherwise the result is inconclusive. The prospective prediction is directional: repair
112 at the diagnosed interface should improve over the baseline and outperform a repair at a non-dominant
113 layer. All states, including harmed cases, remain in the analysis.

114 3.4 Fixed traces versus adaptive search

115 On a frozen candidate set, substituting a score isolates ranking quality. Adaptive CEM is different:

$$q \longrightarrow \text{elite selection} \longrightarrow \text{proposal update} \longrightarrow \text{future candidate distribution}.$$

116 Changing the score therefore changes both ranking and search coverage after the first update. Fixed-
117 trace headroom and end-to-end adaptive-search gain answer different questions; equality between
118 them is neither assumed nor required.

119 4 Experimental Setup

120 4.1 Controlled local Push-T

121 Push-T is a contact-rich planar manipulation task [Florence et al., 2022]. The controlled local protocol
122 uses the frozen action-conditioned JEPA world model and 100 fresh cloned anchor states. Each
123 state has an identical 81-action local angular reference family at horizon H6; a finite-grid CEM-like
124 planner receives 65 queries. The frozen action-blind ridge readout is trained on development states
125 from ground-truth latent transitions and simulator costs. Exact state restoration, action alignment,
126 temporal indexing, and deterministic replay were validated before attribution. This restricted local
127 action slice enables auditable candidate matching; it is *not* the official Push-T planner.

128 The diagnosis, metric repair, and inference-precision control were fixed before repair computation.
129 The repair replaces latent L2 with the frozen readout under the same query budget. The control
130 changes predictor inference from BF16 to FP32 while retaining latent L2; it tests numerical precision
131 only, not a retrained or architecturally improved predictor.

132 4.2 Independent Wall replication

133 Wall uses the frozen environment/model interface and a controlled one-dimensional waypoint can-
134 didate family. Development (60), diagnosis (80), and repair-validation (80) states have disjoint
135 seeds, cloned states, and layouts. The protocol was frozen before diagnosis, and the repair choice
136 was recorded while the repair cache was absent. The pre-specified predictor repair is fit only on
137 development data and evaluated on 80 entirely new states. A metric repair serves as the wrong-layer
138 control. Exact replay, physical labels, cost monotonicity, and rollout alignment were audited before
139 scaling.

140 4.3 Official high-dimensional Push-T H6

141 The frozen H6 validation uses the official one-shot CEM setting: 30 iterations, 300 sequences
142 per iteration, 10 elites, 9,000 model queries, and the returned final mean. Each H6 model action
143 contains five 2-D simulator controls, so CEM searches 60 scalars and the plan executes 30 controls.
144 Thirty fresh near-contact initial states use contiguous precommitted seeds with no filtering, retry, or
145 replacement.

146 Every sequence queried by baseline latent-L2 CEM is retained and simulator-evaluated. GT-readout,
147 predicted-readout, and predicted-L2 rankings on this identical 9,001-sequence trace per episode
148 isolate fixed-trace prediction and metric headroom. A matched-budget CEM run changes only the
149 scorer to the frozen readout. Because this changes elite updates, the two adaptive traces are not
150 candidate matched; their union is used only as a finite audit reference, never as a global action oracle.

Table 1: Frozen cross-regime results. Gaps and repair effects are mean normalized-regret contrasts with 95% state/episode-bootstrap CIs. Positive repair values are improvements. H6 is not a validated repair because its interval crosses zero.

Regime	Prediction gap	Metric gap	Diagnosis	Targeted intervention result
Local Push-T	0.0071 [−0.0051, 0.0198]	0.0851 [0.0608, 0.1109]	Decision metric	Metric scorer: +0.0815 [0.0563, 0.1079]
Wall	0.5636 [0.4871, 0.6373]	−0.0279 CI crosses 0	Prediction	Predictor repair: +0.3891 [0.2930, 0.4804]
Push-T H6	0.1608 [0.0565, 0.2725]	0.0950 [0.0026, 0.1978]	Mixed; prediction > metric	Scorer: +0.0843 [−0.00008, 0.1684]; inconclusive

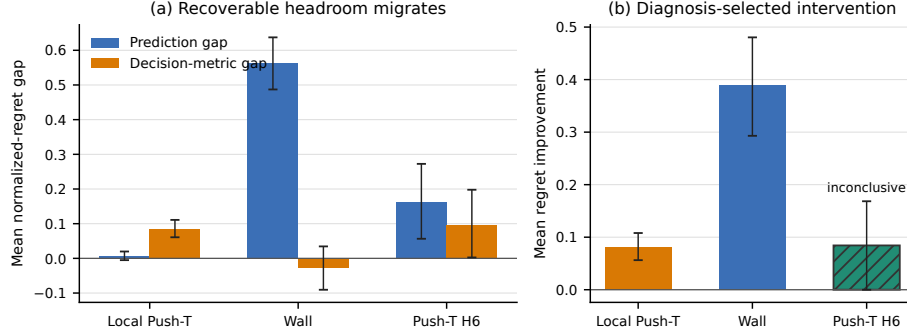


Figure 2: **Across-regime diagnosis and intervention.** (a) Candidate-matched prediction and decision-metric headroom. (b) Mean regret improvement from the diagnosis-selected intervention. Error bars are 95% paired bootstrap CIs over states or episodes. The hatched H6 result is directionally positive but statistically inconclusive because its CI crosses zero.

151 **Statistics.** Reported intervals are 95% paired nonparametric bootstrap confidence intervals over
152 independent anchor states or episodes. No candidate-level pseudoreplication is used. Full protocol
153 and artifact details appear in Appendix A.

154 5 Results

155 Table 1 and Figure 2 summarize the frozen results. The dominant recoverable interface changes
156 across the three planning regimes.

157 5.1 Local Push-T: a decision-metric bottleneck

158 The ground-truth-latent and predicted-latent task readouts achieve state-level Spearman correlations
159 of 0.696 and 0.689 with simulator cost; their normalized regrets are 0.0289 and 0.0360. Consequently,
160 prediction headroom is only 0.0071 (95% CI [−0.0051, 0.0198]). Under the same predicted futures,
161 replacing the readout with latent L2 exposes a much larger metric gap of 0.0851 [0.0608, 0.1109].
162 Proposal headroom is 0.0074. Under the frozen rule, this controlled local protocol is decision-metric
163 limited.

164 The diagnosis-selected repair reduces regret from 0.1237 with BF16 prediction and latent L2 to 0.0421
165 with the task readout: an improvement of 0.0815 [0.0563, 0.1079]. Across states, 62% improve, 21%
166 are unchanged, and 17% are harmed. The FP32-predictor plus latent-L2 control improves regret by
167 only 0.0017, with a CI crossing zero. This supports the narrow statement that increasing predictor
168 inference precision did not recover planning performance; FP32 is not evidence that every possible
169 predictor improvement would fail.

170 The observed repair closely matches the prior metric gap, whereas the precision control aligns with
171 the negligible prediction gap. This diagnosis-to-intervention agreement is the principal local Push-T
172 result.

5.2 Wall: an independent prediction bottleneck

On 80 diagnosis states, prediction headroom is 0.5636 [0.4871, 0.6373], metric headroom is -0.0279 with a CI crossing zero, and proposal headroom is 0.0333. The same rule that diagnoses local Push-T as metric limited therefore diagnoses Wall as prediction limited. This counter-regime makes clear that the procedure is not constructed to declare latent L2 the bottleneck.

On 80 entirely new repair states, the pre-specified predictor repair reduces regret from 0.5561 to 0.1670, an improvement of 0.3891 [0.2930, 0.4804]. The wrong-layer metric repair improves regret by 0.0069 [-0.0489 , 0.0625]. Full-trajectory latent MSE falls from 0.8564 to 0.1999.

Average prediction improvement is not sufficient for state-wise decision improvement. In a retained counterexample, one anchor’s latent MSE improves while regret worsens from 0.215 to 0.991. This does not negate the aggregate repair; it bounds the claim: lower predictive error does not guarantee better action selection for every state. The Wall predictor repair uses one training seed, which limits claims about training variability.

5.3 H6: bottleneck migration and metric–search coupling

On the official H6 baseline trace, GT-readout, predicted-readout, and predicted-L2 mean regrets are 0.1724, 0.3332, and 0.4282. Prediction headroom is therefore 0.1608 [0.0565, 0.2725], exceeding metric headroom 0.0950 [0.0026, 0.1978]. The result is mixed—both gaps are positive—but prediction is the larger limitation under the frozen rule. This differs from the controlled local action slice: long 60-D sequences broaden the candidate family and accumulate model error.

Matched-budget baseline and task-readout CEM obtain regrets of 0.4974 and 0.4131. The scorer-repair improvement is 0.0843 [-0.00008 , 0.1684]: 16 episodes improve, 9 are unchanged, and 5 worsen. Because the interval narrowly crosses zero, the result is **directionally positive but statistically inconclusive**, not a validated successful repair.

The adaptive traces explain why fixed-candidate attribution need not predict end-to-end gain exactly. Changing the scorer reduces within-trace selection gap but can redirect future proposals. In a retained H6 episode, the readout selects the simulator-best sequence in its own trace, yet search coverage degrades and normalized regret worsens from 0.4044 to 1.0000. Thus a better ranking rule on available candidates need not expose better candidates under adaptive search.

6 Discussion and Limitations

What the method adds. PhysPlan is more than an ablation table because each contrast is tied to a specific recoverable intervention, evaluated on identical candidates, and followed by a pre-specified targeted repair and wrong-layer control in the controlled regimes. It does not uniquely assign causal responsibility: gaps are practical ceilings under explicit substitutions, and components interact. The value is prospective localization before engineering effort, not a claim of an additive causal factorization.

Why not always improve prediction? Prediction is clearly limiting on Wall and becomes the larger fixed-trace limitation in H6. It is not the dominant recoverable interface in controlled local Push-T, where a simple metric repair recovers substantial regret and higher predictor inference precision does not. The conclusion is therefore neither “prediction does not matter” nor “metrics are generally limiting,” but that the diagnosis must be repeated when the task, action family, or planner changes.

Scope. Controlled local Push-T uses a restricted local action slice, and Wall uses a controlled 1-D waypoint construction. These designs strengthen candidate-matched attribution but limit behavioral breadth. H6 supplies a realistic high-dimensional planning test, yet contains only 30 episodes; the scorer-repair CI narrowly crosses zero. Wall’s predictor repair has one training seed. The local Push-T repair shares final states with diagnosis, although its intervention was committed before computation; Wall alone uses a disjoint repair set.

Oracle and interaction limits. The GT-readout condition combines representation observability with the finite capacity and development distribution of the ridge readout; it is a practical ceiling, not

221 a pure representation oracle. Oracle gaps need not be additive. Under CEM, score, elite choice, and
222 proposal coverage interact, so fixed-trace metric headroom is not equivalent to an end-to-end scorer
223 intervention. Cross-task gap magnitudes also use task-specific finite-set normalization and should not
224 be interpreted as a common physical scale.

225 **Generality and reliability.** The study covers one latent world-model family and two simulated
226 tasks. It does not establish broad validity for real robots, vision–language–action models, learned
227 policies, or other planners. Average gains conceal heterogeneous states: both controlled repairs harm
228 some cases, and lower latent MSE does not guarantee state-wise decision improvement. Deployment
229 would require uncertainty-aware gating or fallback mechanisms, which are outside this frozen study.

230 7 Conclusion

231 Planning through a latent world model is a multi-interface decision pipeline. Controlled oracle
232 substitutions identify decision-metric headroom in a local Push-T action family and prediction
233 headroom in an independent Wall replication; pre-specified repairs recover the diagnosed gaps in
234 both controlled regimes. Under the official Push-T H6 CEM planner, prediction becomes the larger
235 limitation and rescoring changes search coverage, so the scorer intervention is directionally positive
236 but inconclusive. The supported conclusion is methodological and bounded: the dominant recoverable
237 bottleneck can migrate with the planning regime. Diagnose before retraining the world model or
238 changing the planner.

References

- Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zhohus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khalidov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning, 2025. URL <https://arxiv.org/abs/2506.09985>.
- Alex Ayoub, Zeyu Jia, Csaba Szepesvari, Mengdi Wang, and Lin Yang. Model-based reinforcement learning with value-targeted regression. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 463–474. PMLR, 2020. URL <https://proceedings.mlr.press/v119/ayoub20a.html>.
- Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/hash/3de568f8597b94bda53149c7d7f5958c-Abstract.html.
- Amir-Massoud Farahmand, Andre Barreto, and Daniel Nikovski. Value-aware loss function for model-based reinforcement learning. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pages 1486–1494. PMLR, 2017. URL <https://proceedings.mlr.press/v54/farahmand17a.html>.
- Pete Florence, Corey Lynch, Andy Zeng, Oscar A Ramirez, Ayzaan Wahid, Laura Downs, Adrian Wong, Johnny Lee, Igor Mordatch, and Jonathan Tompson. Implicit behavioral cloning. In *Proceedings of the 5th Conference on Robot Learning*, volume 164 of *Proceedings of Machine Learning Research*, pages 158–168. PMLR, 2022. URL <https://proceedings.mlr.press/v164/florence22a.html>.
- Christopher Grimm, Andre Barreto, Satinder Singh, and David Silver. The value equivalence principle for model-based reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 33, pages 5541–5552, 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/hash/3bb585ea00014b0e3ebe4c6dd165a358-Abstract.html.
- Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2555–2565. PMLR, 2019. URL <https://proceedings.mlr.press/v97/hafner19a.html>.
- Nick Hansen, Hao Su, and Xiaolong Wang. TD-MPC2: Scalable, robust world models for continuous control. In *International Conference on Learning Representations*, pages 47376–47405, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/cf73d57b6dcda32b293df7c2d5341f49-Abstract-Conference.html.
- Reuven Y. Rubinstein. The cross-entropy method for combinatorial and continuous optimization. *Methodology and Computing in Applied Probability*, 1:127–190, 1999. doi: 10.1023/A:1010091220143.
- Basile Terver, Tsung-Yen Yang, Jean Ponce, Adrien Bardes, and Yann LeCun. What drives success in physical planning with joint-embedding predictive world models?, 2026. URL <https://arxiv.org/abs/2512.24497>.
- Claas A Voelcker, Anastasiia Pedan, Arash Ahmadian, Romina Abachi, Igor Gilitschenski, and Amir-Massoud Farahmand. Calibrated value-aware model learning with probabilistic environment models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 61745–61768. PMLR, 2025. URL <https://proceedings.mlr.press/v267/voelcker25a.html>.

288 Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World models on pre-trained
289 visual features enable zero-shot planning. In *Proceedings of the 42nd International Conference on*
290 *Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 79115–79135.
291 PMLR, 2025. URL <https://proceedings.mlr.press/v267/zhou25t.html>.

A Detailed Protocol and Reproducibility

A.1 Frozen experimental record

The manuscript is produced from a completed and frozen experimental record; no experiments were rerun for paper preparation. The final code state is commit ceb6122, and the official H6 protocol was frozen at commit 3cbd3c5. Repository validation records confirm 30/30 paired H6 JSON/NPZ traces, 18/18 related tests, 53/53 immutable historical scientific artifacts, and 117/117 SHA-256 manifest checks. Commit hashes are provided without an identifying repository URL to preserve anonymity.

The authoritative machine-readable summaries are:

- controlled local Push-T diagnosis and repair summaries;
- Wall diagnosis and repair summaries, with separate development, diagnosis, and repair-state manifests; and
- the official H6 aggregate summary plus all 30 paired per-episode raw traces.

Each headline value in the paper is mapped to its report section and saved artifact in the anonymized supplementary evidence audit.

A.2 Candidate sets and intervention controls

In controlled local Push-T, every condition uses the same 81-action reference family and the same 65-query budget per anchor. In Wall, all readout comparisons within a diagnosis state use the same one-dimensional waypoint candidates. For fixed-trace H6 attribution, all scores rank the identical 9,001 sequences retained from the baseline trace. Consequently, each fixed-trace regret contrast measures selection on matched candidates rather than proposal quality.

The adaptive H6 comparison is deliberately labeled differently. Baseline and readout CEM share horizon, population, elite count, query budget, initialization, and random-seed policy, but their candidate traces diverge once scores change the elite set. The trace union is a finite diagnostic reference, not an oracle over the continuous action space.

A.3 Statistical unit and confidence intervals

Local Push-T and Wall bootstrap independent anchor states; H6 bootstraps episodes. Candidate actions and repeated queries within a state are dependent and are not resampled as independent trials. All reported confidence intervals are paired 95% nonparametric bootstrap intervals from the frozen analyses. The repair accounting retains every improved, unchanged, and harmed state or episode.

B Additional Frozen Results

Table 2: Supporting frozen measurements omitted from the compact main table. Lower regret and MSE are better.

Regime	Quantity	Value
Local Push-T	GT / predicted readout Spearman ρ	0.696 / 0.689
Local Push-T	GT / predicted readout regret	0.0289 / 0.0360
Local Push-T	FP32 predictor + L2 improvement	0.0017 (CI crosses 0)
Local Push-T	repair outcomes	62% / 21% / 17% improved / same / harmed
Wall	baseline / predictor-repair regret	0.5561 / 0.1670
Wall	wrong-layer metric repair	0.0069 [−0.0489, 0.0625]
Wall	full-trajectory latent MSE	0.8564 $\rightarrow$ 0.1999
H6	GT-readout / pred-readout / pred-L2 regret	0.1724 / 0.3332 / 0.4282
H6	baseline / scorer-repair regret	0.4974 / 0.4131
H6	repair outcomes	16 / 9 / 5 improved / same / harmed

323 **C Claim Boundaries**

324 The evidence does not support a universal claim that latent planners are prediction limited or metric
325 limited. It does not make oracle gaps additive or uniquely causal, and the FP32 control does not
326 test a general predictor improvement. The official H6 scorer repair is not statistically significant
327 under the frozen criterion. The controlled local Push-T protocol is not the official planner. Finally,
328 the experiments do not establish transfer to real robots, learned policies, or vision–language–action
329 systems.